



THE
KING'S
SCHOOL

2021

Year 12
Alternative Higher School Certificate Trial Examination

Mathematics Extension 1

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A Reference Sheet is provided
- In Questions 11–14, show relevant mathematical reasoning and/or calculations

Total marks: 70

Section I 10 marks

- Attempt Questions 1–10
- Use the Multiple Choice answer sheet provided
- Allow about 15 minutes for this section

Section II 60 marks

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section

Examiners' Use Only						Setter/Checker: External/GJD	
Question	Calculus	Functions	Combinatorics	Statistical Analysis	Trigonometric Functions	Vectors	Total
Section I							
1-10	4, 6, 9, 10 / 4	2, 3, 7 / 3	8 / 1		1 / 1	5 / 1	/10
Section II							
11		a, b / 5	d / 4		c / 2	e / 4	/15
12	d / 3	a, e / 5		c / 3		b / 4	/15
13	a, b, d / 9				c / 6		/15
14		c 6 a / 4				b / 5	/15
Total	/16	/19	/9	/3	/9	/14	/70

Section 1

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10.

1 Which of the following expressions is equivalent to $3\cos x - \sqrt{3}\sin x$?

(A) $2\sqrt{3}\cos\left(x + \frac{\pi}{3}\right)$

(B) $2\sqrt{3}\cos\left(x - \frac{\pi}{3}\right)$

(C) $2\sqrt{3}\cos\left(x + \frac{\pi}{6}\right)$

(D) $2\sqrt{3}\cos\left(x - \frac{\pi}{6}\right)$

2 Which of the following is the correct expansion of $(2x - 1)^3$?

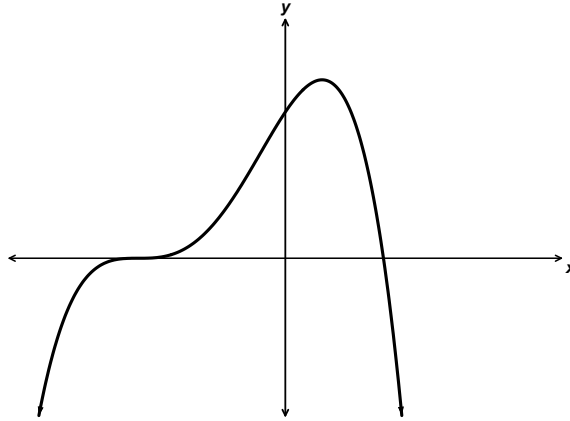
(A) $8x^3 + 12x^2 + 6x - 1$

(B) $8x^3 - 12x^2 + 6x - 1$

(C) $8x^3 - 4x^2 + 2x - 1$

(D) $2x^3 - 12x^2 + 6x - 1$

- 3 The graph of a polynomial $y = P(x)$ is shown below.

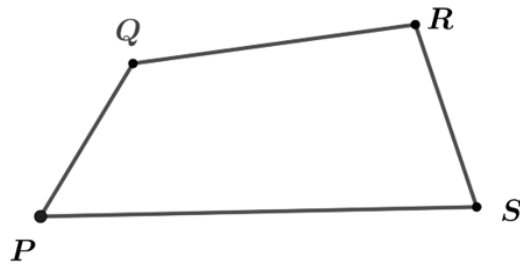


What is the minimum possible degree of the polynomial $[P(x)]^2$?

- (A) 2
- (B) 4
- (C) 8
- (D) 16
- 4 Which of the following is the derivative of $y = x^2 \sin^{-1}(2x)$?

- (A) $2x \sin^{-1}(2x) + \frac{2x^2}{\sqrt{1-4x^2}}$
- (B) $2x \sin^{-1}(2x) + \frac{x^2}{\sqrt{1-4x^2}}$
- (C) $2x \sin^{-1}(2x) + \frac{2x^2}{\sqrt{1-2x^2}}$
- (D) $2x \sin^{-1}(2x) + \frac{x^2}{\sqrt{1-2x^2}}$

- 5 $PQRS$ is a quadrilateral where $\overrightarrow{PQ} = \vec{p}$, $\overrightarrow{QR} = \vec{q}$, $\overrightarrow{RS} = \vec{r}$ and $\overrightarrow{SP} = \vec{s}$



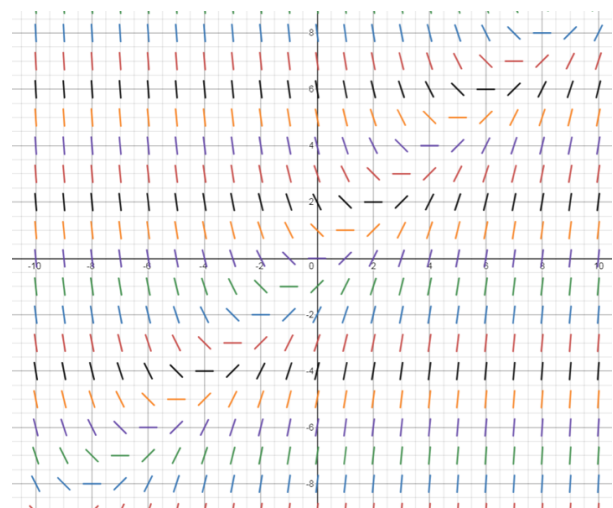
Which of the following is true?

- (A) $\vec{p} + \vec{q} = \vec{s} + \vec{r}$
- (B) $\vec{p} + \vec{q} = -(\vec{s} + \vec{r})$
- (C) $\vec{p} + \vec{s} = \vec{q} + \vec{r}$
- (D) $\vec{p} + \vec{s} = -(\vec{q} - \vec{r})$

- 6 The slope field for a first order differential equation is shown below.

Which of the following could be the differential equation represented?

- (A) $\frac{dy}{dx} = x + y$
- (B) $\frac{dy}{dx} = x - y$
- (C) $\frac{dy}{dx} = y - x$
- (D) $\frac{dy}{dx} = -xy$



7 Which of the following is the solution set of $\frac{2}{1-3^x} > -1$?

- (A) $x < 0$
- (B) $x < 1$
- (C) $0 < x < 1$
- (D) $x < 0$ or $x > 1$

8 A bag contains 5 identical blue marbles, 6 identical black marbles and 3 identical red marbles. Three marbles are drawn at random.

Which expression below gives the correct probability that exactly two blue marbles are drawn?

- (A) $\frac{{}^5C_2}{{}^{14}C_3}$
- (B) $\frac{{}^5C_2 \times {}^9C_1}{{}^{14}C_3}$
- (C) $\frac{2}{5} \times \frac{1}{9}$
- (D) $\frac{2}{14} \times \frac{1}{9}$

9 Which of the following is the value of $\int_0^{\frac{1}{2}} \frac{1}{\sqrt{4-4x^2}} dx$?

(A) $\frac{1}{4}$

(B) $\frac{\pi}{12}$

(C) $\frac{\pi}{6}$

(D) 1

10 The population, P , of animals, in an environment in which there are scarce resources, is increasing such that $\frac{dP}{dt} = P(100 - P)$, where t is time.

The initial population is 20 animals.

Which of the following is true?

(A) $P = 100 - 80e^{100t}$

(B) The population is increasing most rapidly when $P = 50$.

(C) The population is increasing most rapidly when $t = 50$.

(D) The maximum population is $P = 50$.

Section II

60 marks

Attempt Questions 11 – 14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

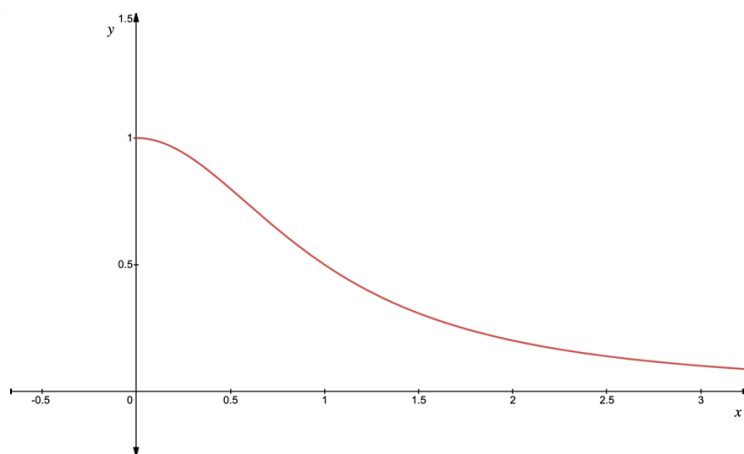
Question 11 (15 marks) Use a SEPARATE writing booklet.

(a) Given $P(x) = x^3 - 5x^2 + 8x - 4$,

(i) Find $P(1)$. 1

(ii) Factorise $P(x)$ 2

(b) 2



The diagram above shows the function $f(x) = \frac{1}{x^2 + 1}$ for $x \geq 0$.

State the domain and range of the inverse function $f^{-1}(x)$.

(c) Given $\cos x = -\frac{1}{3}$ and $\sin x > 0$ find the exact value of $\sin 2x$. 2

Question 11 continues on page 8

Question 11 (continued)

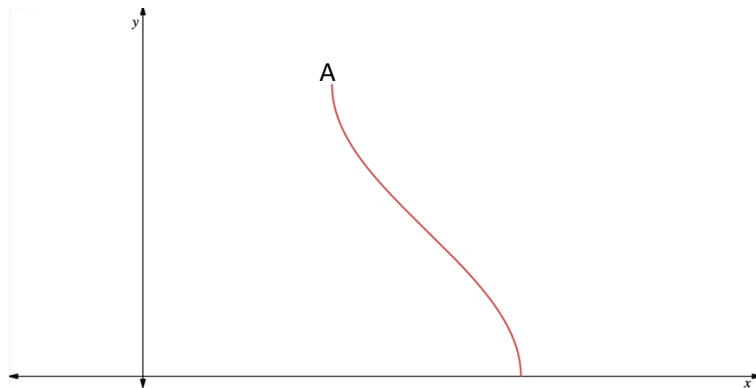
- (d) Ten members of a sporting team have the numbers 1 to 10 on the back of their jumpers.
- (i) Explain why, if six players are randomly chosen, there must be at least one pair of players whose numbers add to 11. 2
- (ii) What is the minimum number of players that need to be chosen to ensure that at least one pair of players have numbers that add to 12? 2
- (e) A projectile is fired from horizontal ground such that horizontal and vertical displacements (in metres) from the point of projection, at time t seconds, are given by $x = (30\sqrt{3})t$ and $y = 30t - 5t^2$.
- (i) Find the angle at which the projectile was fired. 3
- (ii) Find the speed at which the projectile was fired. 1

Question 12 on page 9

Question 12 (15 marks) Use a SEPARATE writing booklet.

(a)

2



The graph above shows $y = 2 \cos^{-1}(2x - 3)$.

Find the coordinates of point A .

- (b) (i) The vectors $\vec{p} = \begin{pmatrix} a \\ 3 \end{pmatrix}$ and $\vec{q} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$ are parallel. 1

Find a .

- (ii) The vector $\vec{r} = \begin{pmatrix} x \\ y \end{pmatrix}$ is perpendicular to vector $\vec{q}$ and 3

$$|\vec{q} - \vec{r}| = \sqrt{65}. \text{ Find the possible values of } x \text{ and } y.$$

- (c) The probability that a certain drug cures a disease is 0.8. The drug is administered to 10 000 patients.

Let X be the binomial random variable representing the number of patients cured.

- (i) Find $E(X)$. 1
- (ii) Show that X has a standard deviation of 40. 1
- (iii) Use a normal approximation to find the probability that X is less than 8040. 1

Question 12 continues on page 10

Question 12 (continued)

- (d) A curve has gradient given by $\frac{dy}{dx} = 2xy$. Find an expression for **3**

y in terms of x given that the curve passes through the point $(0, 4)$

- (e) Use mathematical induction to prove **3**

$$\frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \dots + \frac{n}{2^n} = 2 - \frac{n+2}{2^n} \quad \text{for integers } n \geq 1$$

Question 13 on page 11

Question 13 (15 marks) Use a SEPARATE writing booklet.

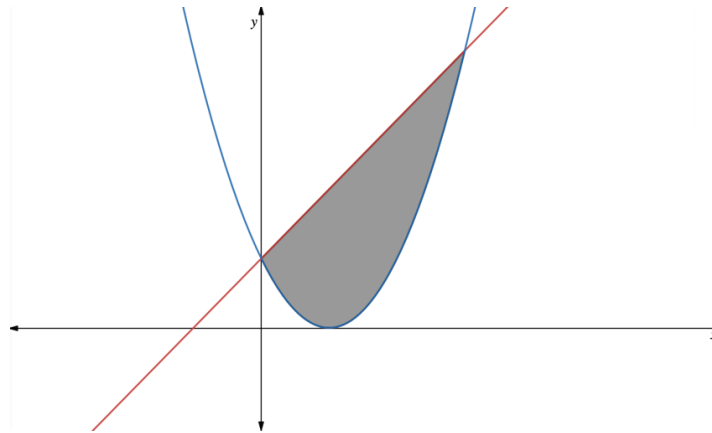
(a) Find $\int_0^{\frac{\pi}{2}} \sin^2 x \cos x \, dx$ 2

(b) Use the substitution $x = 2 \sin \theta$ to find $\int_0^1 \sqrt{4-x^2} \, dx$. 3

(c) (i) Show $\sin(x-y)\sin(x+y) = \sin^2 x - \sin^2 y$. 3

(ii) Solve $\sin^2 3x - \sin^2 x = \sin 4x$ for $0 \leq x \leq \pi$ 3

(d) The shaded area in the diagram below is bounded by $y = (x-1)^2$ 4
and $y = x+1$. The area is rotated around the x -axis to form a solid.
Find the volume of the solid generated.



Question 14 on page12

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) Use the identity

$$(1+x)^{n+m} = (1+x)^n (1+x)^m$$

to show that, if $m > n$,

2

$$\binom{n}{0} \binom{m}{n} + \binom{n}{1} \binom{m}{n-1} + \binom{n}{2} \binom{m}{n-2} + \cdots + \binom{n}{n} \binom{m}{0} = \binom{m+n}{n}$$

- (ii) Simon has n white canaries and m yellow canaries, where $m > n$.

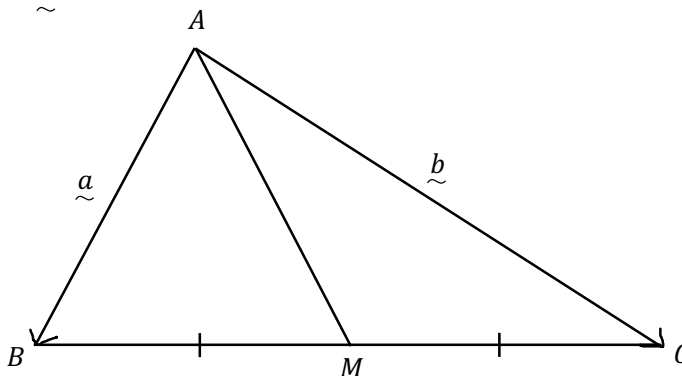
2

He wants to move n canaries, with at least one of each colour, to a new cage.

What is the number of ways this could be done?

- (b) In the diagram, M is the midpoint of the side BC of triangle ABC .

Let $\overrightarrow{AB} = \underline{a}$ and $\overrightarrow{AC} = \underline{b}$.



Copy the diagram.

- (i) Find the vectors $\overrightarrow{AM}$ and $\overrightarrow{CM}$ in terms of $\underline{a}$ and $\underline{b}$.

2

- (ii) Show that $|\overrightarrow{AM}|^2 + |\overrightarrow{CM}|^2 = \frac{1}{2} (|\overrightarrow{AB}|^2 + |\overrightarrow{AC}|^2)$

3

- (c) (i) Show that $8 \cos^4 \theta - 8 \cos^2 \theta + 1 - \cos 4\theta = 0$.

2

- (ii) By letting $x = \cos \theta$ in the equation $16x^4 - 16x^2 + 2 - \sqrt{2} = 0$,

2

show that $\cos 4\theta = \frac{\sqrt{2}}{2}$.

- (iii) Prove that $\cos^2 \frac{\pi}{16} \cos^2 \frac{7\pi}{16} = \frac{2 - \sqrt{2}}{16}$.

2

End of Examination

SUGGESTED SOLUTIONS 2021 Mathematics Extension 1 Trial HSC Examination

Section 1

10 marks

Questions 1 – 10 (1 mark each)

Question 1 (1 mark)

Outcomes assessed: ME11-3

Targeted Performance Bands: E2

Solution	Answer	Mark
$2\sqrt{3} \cos\left(x + \frac{\pi}{6}\right) = 2\sqrt{3} \left(\cos x \cos \frac{\pi}{6} - \sin x \sin \frac{\pi}{6} \right)$ $= 2\sqrt{3} \left(\frac{\sqrt{3}}{2} \cos x - \frac{1}{2} \sin x \right)$ $= 3 \cos x - \sqrt{3} \sin x$	C	1

Question 2 (1 mark)

Outcomes assessed: ME11-2

Targeted Performance Bands: E2-E3

Solution	Answer	Mark
$(2x - 1)^3 = (2x + (-1))^3$ $= (2x)^3 + 3(2x)^2(-1) + 3(2x)(-1)^2 + (-1)^3$ $= 8x^3 - 12x^2 + 6x - 1$	B	1

Question 3 (1 mark)

Outcomes Assessed: ME11-2

Targeted Performance Bands: E2-E3

Solution	Answer	Mark
Minimum degree of $P(x)$ is 4 $\therefore$ Minimum degree of $[P(x)]^2$ is $2 \times 4 = 8$	C	1

Question 4 (1 mark)**Outcomes assessed:** ME12-1**Targeted Performance Bands:** E2

Solution	Answer	Mark
$y = x^2 \sin^{-1}(2x)$ $\frac{dy}{dx} = 2x(\sin^{-1}(2x)) + x^2 \left(\frac{1}{\sqrt{1-(2x)^2}} \times 2 \right)$ $= 2x \sin^{-1}(2x) + \frac{2x^2}{\sqrt{1-4x^2}}$	A	1

Question 5 (1 mark)**Outcomes assessed:** ME12-2**Targeted Performance Bands:** E3

Solution	Answer	Mark
$\vec{p} + \vec{q} + \vec{r} + \vec{s} = 0$ $\vec{p} + \vec{q} = -(\vec{s} + \vec{r})$	B	1

Question 6 (1 mark)**Outcomes assessed:** ME12-1, ME12-4**Targeted Performance Bands:** E3

Solution	Answer	Mark
	B	1

Question 7 (1 mark)**Outcomes Assessed: ME11-2****Targeted Performance Bands: E4**

Solution	Answer	Mark
$\frac{2}{1-3^x} > -1$ Let $u = 1-3^x$ $\frac{2}{u} > -1$ multiply both sides by u^2 $2u > -u^2$ $2u + u^2 > 0$ $u(2+u) > 0$ $u < -2$ or $u > 0$ $1-3^x < -2$ or $1-3^x > 0$ $1-3^x < -2 \Rightarrow 3^x > 3$ $1-3^x > 0 \Rightarrow 3^x < 1$ $\therefore x < 0$ or $x > 1$	D	1

Question 8 (1 mark)**Outcomes Assessed: ME11-5****Targeted Performance Bands: E3**

Solution	Answer	Mark
2 blue marbles chosen in 5C_2 ways. One non-blue marble chosen in 9C_1 ways. 3 Marbles chosen in ${}^{14}C_3$ ways. $\therefore \text{Prob}(\text{exactly 2 blue marbles}) = \frac{{}^5C_2 \times {}^9C_1}{{}^{14}C_3}$	B	1

Question 9 (1 mark)**Outcomes Assessed: ME12-1****Targeted Performance Bands: E3-E4**

Solution	Answer	Mark
$\int_0^{\frac{1}{2}} \frac{1}{\sqrt{4-4x^2}} dx = \frac{1}{2} \int_0^{\frac{1}{2}} \frac{1}{\sqrt{1-x^2}} dx$ $= \frac{1}{2} \left[\sin^{-1} x \right]_0^{\frac{1}{2}}$ $= \frac{1}{2} \left[\frac{\pi}{6} - 0 \right]$ $= \frac{\pi}{12}$	B	1

Question 10 (1 mark)**Outcomes assessed: ME11-4****Targeted Performance Bands: E4**

Solution	Answer	Mark
$\frac{dP}{dt}$ has maximum value when $P = 50$ $\therefore$ population is increasing most rapidly when $P = 50$.	B	1

Question 11 (15 marks)**(a) (i) (1 mark)****Outcomes assessed: ME11-1****Targeted Performance Bands: E2**

Criteria	Marks
Correctly shows the result.	1

Sample answer:

$$P(1) = 1 - 5 + 8 - 4 = 0$$

(a) (ii) (2 marks)

Outcomes assessed: ME11-1

Targeted Performance Bands: E2

Criteria	Marks
• Correct factorisation.	2
• Recognises that $(x-1)$ is a factor of $P(x)$ and makes some attempt to factorise.	1

Sample answer:

$$\begin{array}{l} \frac{x^2 - 4x + 4}{(x-1) \overline{) x^3 - 5x^2 + 8x - 4}} \\ P(x) = (x-1)(x-2)^2 \end{array}$$

(b) (2 marks)

Outcomes assessed: ME11-3

Targeted Performance Bands: E3

Criteria	Marks
• Correct Range	2
• Correct Domain.	1

Sample answer:

$$\begin{array}{l} D_{f^{-1}} = R_f \Rightarrow D_{f^{-1}} : 0 < x \leq 1 \\ R_{f^{-1}} = D_f \Rightarrow R_{f^{-1}} : y \geq 0 \end{array}$$

(c) (2 marks)

Outcomes assessed: ME11-3

Targeted Performance Bands: E3

Criteria	Marks
• Correct answer.	2
• Finds the value of $\sin x$ or equivalent progress.	1

Sample answer:

$$\cos x = -\frac{1}{3}$$

$$\sin^2 x = 1 - \cos^2 x = \frac{8}{9}$$

$$\sin x = \sqrt{\frac{8}{9}} = \frac{2\sqrt{2}}{3} \text{ (given } \sin x > 0 \text{)}$$

$$\sin 2x = 2 \sin x \cos x = 2 \times \left(-\frac{1}{3}\right) \times \frac{2\sqrt{2}}{3} = -\frac{4\sqrt{2}}{9}$$

(d) (i) (2 marks)

Outcomes assessed: ME11-5, ME12-7

Targeted Performance Bands: E3

Criteria	Marks
• Provides correct explanation.	2
• Identifies that there are 5 pairs that give a sum of 11.	1

Sample answer:

Each number has a "partner" to give a total of 11

There are 5 pairs that give a total of 11:

(1,10);(2,9);(3,8);(4,7);(5,6)

That is: 5 pigeonholes

$\therefore$ by the pigeonhole principle, if 6 players are chosen
there will be at least one pair that gives a sum of 11.

(d) (ii) (1 mark)

Outcomes assessed: ME11-5, ME12-7

Targeted Performance Bands: E3

Criteria	Marks
• Correct answer	2
• Recognises that there are some numbers that don't have "pair" to sum to 12.	1

Sample answer:

The numbers "1" and "6" do not have a "partner" to give a total of 12

There are 4 pairs that give a total of 12:

(2,10); (3,9); (4,8); (5,7)

Will need to choose 7 players.

(e) (i) (3 marks)

Outcomes assessed: ME12-2

Targeted Performance Bands: E3-E4

Criteria	Marks
• Correct answer.	3
• Correctly determines initial vertical and horizontal velocities.	2
• Makes some progress (e.g. makes some attempt to find initial velocities).	1

Sample answer:

$$x = 30\sqrt{3}t \Rightarrow \frac{dx}{dt} = 30\sqrt{3}$$

$$y = 30t - 5t^2 \Rightarrow \frac{dy}{dt} = 30 - 10t$$

$$\text{when } t = 0, \frac{dy}{dt} = 30$$

if θ is the angle of projection, then

$$\tan \theta = \frac{30}{30\sqrt{3}} = \frac{1}{\sqrt{3}}$$

$$\theta = 30^\circ$$

(e) (ii) (1 mark)

Outcomes assessed: HE3

Targeted Performance Bands: E3-E4

Criteria	Marks
• Correct answer	1

Sample answer:

If initial speed is V

then

$$V^2 = 30^2 + (30\sqrt{3})^2$$

$$V = 60 \text{ ms}^{-1}$$

Question 12 (15 marks)

(a) (2 marks)

Outcomes Assessed: ME11-3

Targeted Performance Bands: E2-E3

Criteria	Marks
• Correct answer.	2
• Finds correct domain or range or equivalent.	1

Sample answer:

$$\text{For } y = 2 \cos^{-1}(2x - 3)$$

$$D: -1 \leq 2x - 3 \leq 1$$

$$2 \leq 2x \leq 4$$

$$1 \leq x \leq 2$$

$$R: 0 \leq y \leq 2\pi$$

$$\therefore A \text{ is } (1, 2\pi)$$

(b) (i) (1 mark)

Outcomes assessed: ME12-2

Targeted Performance Bands: E2

Criteria	Marks
• Correct answer.	1

Sample answer:

$$\text{Parallel} \Rightarrow \lambda \begin{pmatrix} a \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$$

$$3\lambda = 4 \Rightarrow \lambda = \frac{4}{3}$$

$$\frac{4}{3}a = 2$$

$$a = \frac{3}{2}$$

(b) (ii) (3 marks)

Outcomes assessed: ME12-2

Targeted Performance Bands: E3

Criteria	Marks
• Correct answer	3
• Writes correct expression for $ \vec{q} - \vec{r} $	2
• Recognises that the dot product is zero.	1

Sample answer:

$$\text{Perpendicular} \Rightarrow \lambda \begin{pmatrix} 2 \\ 4 \end{pmatrix} \bullet \begin{pmatrix} x \\ y \end{pmatrix} = 0$$

$$2x + 4y = 0$$

$$x = -2y$$

$$\vec{q} - \vec{r} = \begin{pmatrix} 2-x \\ 4-y \end{pmatrix} = \begin{pmatrix} 2+2y \\ 4-y \end{pmatrix}$$

$$|\vec{q} - \vec{r}| = \sqrt{65} \Rightarrow \sqrt{(2+2y)^2 + (4-y)^2} = \sqrt{65}$$

squaring

$$4 + 8y + 4y^2 + 16 - 8y + y^2 = 65$$

$$5y^2 + 20 = 65$$

$$y^2 = 9$$

$$y = \pm 3$$

$$\therefore x = 6, y = -3 \text{ or } x = -6, y = 3$$

(c) (i) (1 mark)

Outcomes assessed: ME12-5

Targeted Performance Bands: E2

Criteria	Marks
• Correct answer.	1

Sample answer:

$$E(X) = np = 10000 \times 0.8 = 8000$$

(c) (ii) (1 mark)

Outcomes assessed: ME12-5

Targeted Performance Bands: E2

Criteria	Marks
• Correctly shows the result	1

Sample answer:

$$\begin{aligned} \text{Var}(X) &= np(1-p) \\ &= 10000 \times 0.8 \times 0.2 \\ &= 1600 \\ \sigma &= \sqrt{1600} = 40 \end{aligned}$$

(c) (iii) (1 mark)

Outcomes assessed: ME12-5

Targeted Performance Bands: E3

Criteria	Marks
• Correct answer.	1

Sample answer:

$$\begin{aligned}P(X < 8040) &= P(Z < 1) \\&= 0.5 + 0.34 \\&= 0.84\end{aligned}$$

(d) 3 marks

Outcomes assessed: ME12-1, ME12-4

Targeted Performance Bands: E3

Criteria	Marks
• Correct answer.	3
• Correct integration	2
• Separates the differential equation and makes some attempt to integrate.	1

Sample answer:

$$\begin{aligned}\frac{dy}{dx} &= 2xy \\ \frac{1}{y} dy &= 2x dx \\ \int \frac{1}{y} dy &= \int 2x dx \\ \log_e y &= x^2 + c \\ \text{Passes through } (0, 4) &\Rightarrow c = \log_e 4 \\ \log_e y &= x^2 + \log_e 4 \\ \log_e y - \log_e 4 &= x^2 \\ \log_e \frac{y}{4} &= x^2 \\ y &= 4e^{x^2}\end{aligned}$$

(b) (3 marks)

Outcomes assessed: ME12-1; ME12-7

Targeted Performance Bands: E3 – E4

Criteria	Marks
• Correctly proves the result.	3
• Uses the correct assumption step in attempting to prove the statement is true for $n = k + 1$	2
• Correctly proves that the statement is true when $n = 1$.	1

Sample answer:

Step1 Prove true for $n = 1$

$$LHS = \frac{1}{2}$$

$$RHS = 2 - \frac{1+2}{2} = \frac{1}{2}$$

$\therefore$ true for $n = 1$

Step2 Let $n = k$ be a value for which the statement is true

ie

$$\frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \dots + \frac{k}{2^k} = 2 - \frac{k+2}{2^k}$$

Step3 Prove true for $n = k + 1$

i.e prove

$$\frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \dots + \frac{k}{2^k} + \frac{k+1}{2^{k+1}} = 2 - \frac{(k+1)+2}{2^{k+1}} = 2 - \frac{k+3}{2^{k+1}}$$

$$\begin{aligned} LHS &= 2 - \frac{k+2}{2^k} + \frac{k+1}{2^{k+1}} \\ &= 2 - \left(\frac{2(k+2)}{2^{k+1}} - \frac{k+1}{2^{k+1}} \right) \\ &= 2 - \left(\frac{2k+4-k-1}{2^{k+1}} \right) \\ &= 2 - \frac{k+3}{2^{k+1}} \text{ as required.} \end{aligned}$$

$\therefore$ true by mathematical induction

Question 13 (15 marks)

(a) (2 marks)

Outcomes assessed: ME12-1***Targeted Performance Bands: E3***

Criteria	Marks
• Correct answer	2
• Chooses appropriate substitution.	1

Sample answer:

Let $u = \sin x$

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \sin^2 x \cos x \, dx &= \int_0^1 u^2 \, du \\ &= \left[\frac{u^3}{3} \right]_0^1 = \frac{1}{3}\end{aligned}$$

(b) (3 marks)

Outcomes Assessed: ME12-4***Targeted Performance Bands: E4***

Criteria	Marks
• correct answer	3
• correct final integral	2
• any correct attempt	1

Sample answer:

$$x = 2 \sin \theta$$

$$\frac{dx}{d\theta} = 2 \cos \theta$$

$$x = 0 \Rightarrow \theta = 0$$

$$x = 1 \Rightarrow \theta = \frac{\pi}{6}$$

$$\begin{aligned} \int_0^1 \sqrt{4-x^2} \, dx &= \int_0^{\frac{\pi}{6}} \sqrt{4-4\sin^2 \theta} \times 2 \cos \theta \, d\theta \\ &= \int_0^{\frac{\pi}{6}} 2 \cos \theta \times 2 \cos \theta \, d\theta \\ &= 4 \int_0^{\frac{\pi}{6}} \cos^2 \theta \, d\theta \\ &= 4 \int_0^{\frac{\pi}{6}} \frac{1}{2} (1 + \cos 2\theta) \, d\theta \\ &= 2 \left[\theta + \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{6}} \\ &= 2 \left[\frac{\pi}{6} + \frac{1}{2} \sin \frac{\pi}{3} \right]_0^{\frac{\pi}{6}} \\ &= \frac{\pi}{3} + \frac{\sqrt{3}}{2} \end{aligned}$$

(c) (i) (3 marks)

Outcomes Assessed: ME12-3

Targeted Performance Bands: E4

Criteria	Marks
• Correctly shows the result.	3
• Makes some correct use of $\cos 2x$ results or equivalent progress.	2
• Makes some use of “products to sums” results or equivalent progress.	1

Sample answer:

$$\begin{aligned}
\sin(x-y)\sin(x+y) &= \frac{1}{2} \left[\cos[(x-y)-(x+y)] - \cos[(x-y)+(x+y)] \right] \\
&= \frac{1}{2} \left[\cos(-2y) - \cos(2x) \right] \\
&= \frac{1}{2} \left[\cos(2y) - \cos(2x) \right] \\
&= \frac{1}{2} \left[(1 - 2\sin^2 y) - (1 - 2\sin^2 x) \right] \\
&= \sin^2 x - \sin^2 y
\end{aligned}$$

(c) (ii) (3 marks)

Outcomes Assessed: ME12-3

Targeted Performance Bands: E4

Criteria	Marks
• Correctly solves the equation.	3
• Finds one correct non-zero solution.	2
• Correctly uses the result from (i) to transform equation.	1

Sample answer:

$$\begin{aligned}
&\text{From (i) } \sin^2 3x - \sin^2 x = \sin(3x - x)\sin(3x + x) = \sin 2x \sin 4x \\
&\sin 2x \sin 4x = \sin 4x \text{ for } 0 \leq x \leq \pi \\
&\sin 2x \sin 4x - \sin 4x = 0 \\
&\sin 4x(\sin 2x - 1) = 0 \\
&\sin 4x = 0 \Rightarrow 4x = 0, \pi, 2\pi, 3\pi, 4\pi \\
&x = 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi \\
&\sin 2x = 1 \Rightarrow 2x = \frac{\pi}{2} \Rightarrow x = \frac{\pi}{4} \\
&\therefore x = 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi
\end{aligned}$$

(d) (4 marks)

Outcomes Assessed: ME12-4

Targeted Performance Bands: E4

Criteria	Marks
• Provides correct answer.	4
• Makes further progress finding integral.	3
• Writes correct integral expression for the volume.	2
• Finds correct points of intersection	1

Sample answer:

For point of intersection

$$x+1=(x-1)^2$$

$$x+1=x^2-2x+1$$

$$x^2-3x=0$$

$$x=0, x=3$$

$$V=\pi \int_0^3 (x+1)^2 - [(x-1)^2]^2 dx$$

$$=\pi \int_0^3 (x+1)^2 - (x-1)^4 dx$$

$$=\pi \left[\frac{(x+1)^3}{3} - \frac{(x-1)^5}{5} \right]_0^3$$

$$=\pi \left[\left(\frac{64}{3} - \frac{32}{5} \right) - \left(\frac{1}{3} - \frac{1}{5} \right) \right]$$

$$=\frac{74\pi}{5} \text{ units}^3$$

Question 14 (15 marks)

Sample answer:

$$(a) \ i) \ (1+x)^n = \binom{n}{0}x^0 + \binom{n}{1}x^1 + \binom{n}{2}x^2 + \dots + \binom{n}{n-1}x^{n-1} + \binom{n}{n}x^n$$

As $m > n$ this means $(1+x)^m$ has more terms than $(1+x)^n$.

$$(1+x)^m = \binom{m}{0}x^0 + \binom{m}{1}x^1 + \binom{m}{2}x^2 + \dots + \binom{m}{n}x^n \dots \dots + \binom{m}{m-1}x^{m-1} + \binom{m}{m}x^m$$

In the expansion of $(1+x)^n(1+x)^m$, each term in the expansion of $(1+x)^n$ will be multiplied by each term in the expansion of $(1+x)^m$.

To show that the required statement is true, we must equate the coefficients of x^n in the expansions of $(1+x)^n(1+x)^m$ and $(1+x)^{n+m}$

The coefficient of x^n in the expansion of

$(1+x)^n(1+x)^m$ is found by multiplying each term in the expansion of $(1+x)^n$ by the corresponding term in the expansion of $(1+x)^m$, such that the sum of the powers of x is n .

Hence, the corresponding coefficients of x^n will be

$$\binom{n}{0}\binom{m}{n} + \binom{n}{1}\binom{m}{n-1} + \binom{n}{2}\binom{m}{n-2} + \binom{n}{3}\binom{m}{n-3} + \dots + \binom{n}{n-1}\binom{m}{1} + \binom{n}{n}\binom{m}{0}$$

In the expansion of $(1+x)^{n+m}$, the coefficient of

x^n will be $\binom{m+n}{n}$. Hence,

$$\binom{n}{0}\binom{m}{n} + \binom{n}{1}\binom{m}{n-1} + \dots + \binom{n}{n}\binom{m}{0} = \binom{m+n}{n}$$

ii) The number of different combinations possible for n canaries from n white and m yellow canaries is

$$\binom{n}{0}\binom{m}{n} + \binom{n}{1}\binom{m}{n-1} + \dots + \binom{n}{n}\binom{m}{0}$$

From the previous part of this question this would be equivalent to :- $\binom{m+n}{n}$

As Simon must have at least one canary of each colour, no white and n yellow $\binom{n}{0}\binom{m}{n}$, n white and no yellow $\binom{n}{n}\binom{m}{0}$ are not acceptable. Hence, the number of ways Simon could select n canaries, ensuring there was at least one of each colour is

$$\binom{m+n}{n} - \binom{n}{0}\binom{m}{n} - \binom{n}{n}\binom{m}{0} \text{ As } \binom{n}{n} = 1, \binom{m}{0} = 1 \text{ and } \binom{n}{0} = 1 \text{ then}$$

the number of ways is

$$\binom{m+n}{n} - 1 \times \binom{m}{n} - 1 \times 1 = \binom{m+n}{n} - \binom{m}{n} - 1$$

$$b) \ i) \ \overrightarrow{AM} = \overrightarrow{a} + \overrightarrow{BM} \quad (1)$$

also $\overrightarrow{AM} = \overrightarrow{b} + \overrightarrow{CM}$ and as

$$\overrightarrow{CM} = -\overrightarrow{BM} \text{ then}$$

$$\overrightarrow{AM} = \overrightarrow{b} - \overrightarrow{BM} \quad (2)$$

By adding (1) and (2), we get

$$2\overrightarrow{AM} = \overrightarrow{a} + \overrightarrow{b} \text{ that is}$$

$$\overrightarrow{AM} = \frac{1}{2}(\underline{a} + \underline{b})$$

$$\overrightarrow{CM} = -\underline{b} + \overrightarrow{AM}$$

$$\overrightarrow{CM} = -\underline{b} + \frac{1}{2}(\underline{a} + \underline{b})$$

$$\overrightarrow{CM} = -\underline{b} + \frac{1}{2}\underline{a} + \frac{1}{2}\underline{b}$$

$$\overrightarrow{CM} = \frac{1}{2}\underline{a} - \frac{1}{2}\underline{b}$$

$$\overrightarrow{CM} = \frac{1}{2}(\underline{a} - \underline{b})$$

$$\begin{aligned} \text{ii) LHS} &= |\overrightarrow{AM}|^2 + |\overrightarrow{CM}|^2 \\ &= \frac{1}{2}(\underline{a} + \underline{b}) \cdot \frac{1}{2}(\underline{a} + \underline{b}) + \frac{1}{2}(\underline{a} - \underline{b}) \cdot \frac{1}{2}(\underline{a} - \underline{b}) \\ &= \frac{1}{4}(\underline{a} + \underline{b})^2 + \frac{1}{4}(\underline{a} - \underline{b})^2 \\ &= \frac{1}{4}(|\underline{a}|^2 + 2\underline{a}\underline{b} + |\underline{b}|^2 + |\underline{a}|^2 - 2\underline{a}\underline{b} + |\underline{b}|^2) \\ &= \frac{1}{4}(2|\underline{a}|^2 + 2|\underline{b}|^2) \\ &= \frac{1}{2}(|\underline{a}|^2 + |\underline{b}|^2) \\ &= \frac{1}{2}(|\overrightarrow{AB}|^2 + |\overrightarrow{AC}|^2) \\ &= RHS \end{aligned}$$

$$\begin{aligned} \text{c) i) } \cos 4\theta &= 2\cos^2 2\theta - 1 \\ \cos 4\theta &= 2(2\cos^2 \theta - 1)^2 - 1 \\ \cos 4\theta &= 2(4\cos^4 \theta - 4\cos^2 \theta + 1) - 1 \\ \cos 4\theta &= 8\cos^4 \theta - 8\cos^2 \theta + 2 - 1 \\ \cos 4\theta &= 8\cos^4 \theta - 8\cos^2 \theta + 1 \\ 0 &= 8\cos^4 \theta - 8\cos^2 \theta + 1 - \cos 4\theta \end{aligned}$$

$$\begin{aligned} \text{ii) } \cos 4\theta &= 8\cos^4 \theta - 8\cos^2 \theta + 1 \\ 2\cos 4\theta &= 16\cos^4 \theta - 16\cos^2 \theta + 2 \\ \text{By substituting} \\ 16\cos^4 \theta - 16\cos^2 \theta + 2 - \sqrt{2} &= 0 \\ \text{from above, we get} \\ 2\cos 4\theta - \sqrt{2} &= 0 \\ 2\cos 4\theta &= \sqrt{2} \\ \cos 4\theta &= \frac{\sqrt{2}}{2} \end{aligned}$$

$$\begin{aligned} \text{iii) From part ii) the solution to the equation} \\ 16\cos^4 \theta - 16\cos^2 \theta + 2 - \sqrt{2} = 0 \text{ can be found} \\ \text{from } \cos 4\theta &= \frac{\sqrt{2}}{2} \\ \cos 4\theta &= \cos \frac{\pi}{4} \\ 4\theta &= \frac{\pi}{4} + 2n\pi \text{ or } 4\theta = -\frac{\pi}{4} + 2n\pi \end{aligned}$$

$$\theta = \frac{\pi}{16} + \frac{n\pi}{2} \text{ or } \theta = -\frac{\pi}{16} + \frac{n\pi}{2}$$

when $n = 0$, $\theta = \frac{\pi}{16}$ or when $n = 0$, $\theta = -\frac{\pi}{16}$

when $n = 1$, $\theta = \frac{9\pi}{16}$ or when $n = 1$, $\theta = \frac{7\pi}{16}$

when $n = 2$, $\theta = \frac{17\pi}{16}$ or when $n = 2$, $\theta = \frac{15\pi}{16}$

when $n = 3$, $\theta = \frac{25\pi}{16}$ or when $n = 3$, $\theta = \frac{23\pi}{16}$

There is an infinite number of possible values for the angle θ . The four lines above show a few of these possible values.

As the equation $16x^4 - 16x^2 + 2 - \sqrt{2} = 0$ has a degree 4 it should have at maximum four real roots which mean four possible values for $\cos \theta$ regardless of the possible values of θ .

Hence, we need to find four different values for $\cos \theta$ using the possible values for angle θ found above as they are the four roots of this equation.

The four different values of $\cos \theta$ or the four roots of the quartic equation are

$$\cos \frac{\pi}{16}, \cos \frac{15\pi}{16}, \cos \frac{7\pi}{16} \text{ and } \cos \frac{23\pi}{16}$$

The product of these roots is $\frac{2 - \sqrt{2}}{16}$.

This means

$$\cos \frac{\pi}{16} \cos \frac{15\pi}{16} \cos \frac{7\pi}{16} \cos \frac{23\pi}{16} = \frac{2 - \sqrt{2}}{16}$$

Now, as $\cos \frac{15\pi}{16} = -\cos \frac{\pi}{16}$ and $\cos \frac{23\pi}{16} = -\cos \frac{7\pi}{16}$ this means

$$\cos \frac{\pi}{16} \times -\cos \frac{\pi}{16} \cos \frac{7\pi}{16} \times -\cos \frac{7\pi}{16} = \frac{2 - \sqrt{2}}{16}$$

$$\text{Hence, } \cos^2 \frac{\pi}{16} \cos^2 \frac{7\pi}{16} = \frac{2 - \sqrt{2}}{16}$$

Alternative method

From part ii) the solution to the equation

$$16\cos^4 \theta - 16\cos^2 \theta + 2 - \sqrt{2} = 0 \text{ can be found}$$

$$\text{from } \cos 4\theta = \frac{\sqrt{2}}{2}$$

$$\text{So } 4\theta = \frac{\pi}{4}, \frac{7\pi}{4}, \frac{9\pi}{4}, \frac{15\pi}{4} \dots$$

$\therefore \theta = \frac{\pi}{16}, \frac{7\pi}{16}, \frac{9\pi}{16}, \frac{15\pi}{16} \dots$ are solutions to the equation.

$$\text{But } 16\cos^4 \theta - 16\cos^2 \theta + 2 - \sqrt{2} = 0$$

can be reduced to the quadratic

$$16m^2 - 16m + 2 - \sqrt{2} = 0 \text{ where } m = \cos^2 \theta$$

As $\theta = \frac{\pi}{16}, \frac{7\pi}{16}, \frac{9\pi}{16}, \frac{15\pi}{16} \dots$ are solutions to the

initial trig equation, $\cos^2 \frac{\pi}{16}$ and $\cos^2 \frac{7\pi}{16}$ will be roots of this new equation.

The product of the roots of a quadratic equation is

equal to the constant term, divided by the coefficient of m^2 ,

$$\text{Hence, } \cos^2 \frac{\pi}{16} \cdot \cos^2 \frac{7\pi}{16} = \frac{2-\sqrt{2}}{16}$$